

Problem Statement 8

AI-Assisted Medication Adherence Monitoring System

The Challenge

Medication non-adherence is a significant challenge in the management of chronic diseases. It frequently leads to poor health outcomes, increased complications, and higher rates of hospitalization. Healthcare providers often lack continuous visibility into patient medication adherence behavior, making it difficult to identify early signs of non-compliance.

There is a clear need for an assistive AI-based system that can analyze medication adherence data, detect risk patterns, and support patients through awareness and behavioral guidance.

Solution Components

Medication Data Analysis Agent

An AI agent that analyzes medication intake logs, refill history, and missed-dose patterns to understand patient adherence behavior.

Adherence Risk Detection Agent

An AI agent that evaluates adherence trends over time using historical data and predefined rules to identify patients at risk of declining adherence.

Reminder and Lifestyle Support Assistant

An assistive AI agent that provides adherence reminders and educational insights to encourage consistent medication-taking behavior.

This component is non-diagnostic and does not provide medical advice.

Outcome

The system improves medication adherence awareness, enables early identification of non-compliance risks, and supports better long-term treatment outcomes through continuous monitoring and patient engagement.